



Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595

Serving others. Building people. Pursuing excellence.

Customer: DNOW
Customer PO No.: 662682056
TTB Job No.: 25H105093
Code of Fabrication: ASME B16.49-2023

This Data Book contains information proprietary to Tulsa Tube Bending Co.
This information is for the records of the purchaser and their subsidiaries and shall not be commercially distributed.

Induction Bending Data Book Contents

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To:
TULSA TUBE BENDING COMPANY, INC.
DEPT. 1696
TULSA OK 74182
Tel No. 918-446-4461
Fax No. 918-446-1595
Your Vendor Number with us 116550

Please Deliver To:
DNOW L.P.
MT. PLEASANT DSC
1701 GOVER PKWY
MOUNT PLEASANT MI 48858-8140
Tel No: 989-773-9948
Fax No: 989-773-6143

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Information:
PO Date: 03/10/2025
Contact Person: Matthew Sisung
Email: Matthew.Sisung@dnow.com
Currency: USD
Ordering Plant : MI04
Delivery Date: 04/04/2025
Final Destination: USA

***PLEASE SEE
NOTE BELOW***



***"NOTE THIS PO PRINTED OUTPUT IS FOR INFORMATION ONLY.
PO HAS BEEN SUBMITTED ELECTRONICALLY TO VENDOR (TAULIA)"***

INVOICE SUBMISSION & CORRESPONDENCE:
Invoices must be submitted via Taulia to be processed for payment. For information on Taulia, please reach out to vendors-change@dnow.com

Statements/Other inquires s/b directed to:
apresolutionteam@dnow.com

PLEASE SHIP FED EX 36394525

SHIPPING INSTRUCTIONS:
DNOW Shipping Instructions Last updated 06/29/2024
Free freight allowed, use your best method.
If free freight not allowed ensure items are properly packed for transport and follow instructions below.

Palletized Instructions- For single shipments over 150 pounds or multiple shipment greater than a combined weight of 300 pounds please use the link below for carrier selection and BOL. DNOWs PO number must be inputted into the system.

<https://tms.one/DNOW/eTMS.html>

Please contact DNOW Logistics Group if your shipment is over 8ft in length, need to ship more than 6 pallets or if your shipment is over 7,000 pounds. Further instructions will be provided.

Non-Palletized / Parcel Instructions- For single shipments less than 150 lbs. and multiple shipments with a combined weight less than 300 lbs. Ship via UPS.

UPS shipment rules:
(1) Use UPS Account listed below.
(2) Place DNOW PO in the Reference field.
(3) DO NOT PALLETIZE UPS.





Information:	Page 2 of 2
PO Date: 03/10/2025	

(4) Must enter declared shipment value.

Need help please email CorporateLogistics@dnw.com. If this is an emergency call 281-823-4664 or 281-823-4665 for further instructions.

Cost center: 107334
UPS account: 455041

Item	Material/Description	Order Qty.	UM	Price per Unit	Net Value
10	ZFIT INDUCTION BEND 10" X 0.500" WT, X52, INDUCTION BEND 10" X 0.500" WT, X52, 30 DEG, 5R RADIUS ASME B16.49, COATED WITH FBE PER CONSUMERS ENERGYS TECHNICAL SPECIFICATION FOR FBE COATED PIPE ISSUED 1-1-95 LATEST REVISION Sales Order 102575860 Item 000240 Tracking No: 450-000020	8	EA		
20	ZFIT INDUCTION BEND 10" X 0.500" WT, X52 INDUCTION BEND 10" X 0.500" WT, X52, 45 DEG, 5R RADIUS ASME B16.49, COATED WITH FBE PER CONSUMERS ENERGYS TECHNICAL SPECIFICATION FOR FBE COATED PIPE ISSUED 1-1-95 LATEST REVISION Sales Order 102575860 Item 000360 Tracking No: 450-000027	2	EA		

Total Net Item Value excluding TAX USD

St. Louis Pipe & Supply

CONFIDENTIAL



VALLOUREC STAR, LP

Seamless Tubular Products

Certified Test Report
11/28/2023

Order Line #: Q-221125736-1	Description: SEAMLESS, QUENCH AND TEMPERED	SLN (Heat): N90101
Grade: A515L X52Q 46th Edition May 1, 2019	W/O No: DE7158	Lot: N638910
OD (in): 10.750	Wall (in): 0.500	
End Finish: PEB	Lbs/Ft: 54.79	Length: DRL
Comment: Melted and Manufactured in the USA Vallourec Star's QA program meets the req. of DIN EN 10204-2004 Type 3.1 Satisfies NACE MR0175 PAR3.2/ISO 15156-2: 2020 & ANSI/NACE MR0103/ISO 17945:2015 X52Q/LT ASTM A333-18 G1&G6 ASME SA333-2021 G1&G6 ASTM A106B/C-19a ASME SA106B/C-2021 ASTM A53B-22 ASME SA53B-2021 MEETS PSL2		

Mechanical Properties

Orientation	Type	Tensile Size Specimen Cross Section			Strength (ksi)			Elongation	
		Width (in)	Thick (in)	Area (sqin)	Yield	Tensile	Y/T Ratio	Gage Length	%Elong
LONGITUDINAL	STRIP	1.493	0.507	0.7597	65.0	77.8	0.84	2	44.0
LONGITUDINAL	STRIP	1.495	0.483	0.7247	66.5	78.7	0.85	2	43.0
LONGITUDINAL	STRIP	1.494	0.491	0.7362	65.7	77.8	0.84	2	44.0
Calculated Max Internal Yield (psi): 5294									

Supplemental Requirement

Austenitizing Temp	1700 °F
Tempering Temp	1315 °F
Flattening	PASSED
Hardness	SEE ATTACHED
Collapse Min	N/A
Collapse Actual	N/A
Hydrostatic Pressure	3,000 5 SEC HOLD
Other	N/A

Chemical Analysis

Chem Spec: 69Q

NDE SEE ATTACHED

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
HEAT	0.11	1.09	0.012	0.004	0.23	0.15	0.06	0.13	0.02	0.010	0.000	0.044	0.026	0.0014	0.0005	0.003	0.0096	0.34
PRODUCT 1	0.11	1.05	0.011	0.004	0.23	0.14	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0085	0.34
PRODUCT 2	0.11	1.05	0.011	0.004	0.23	0.15	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0086	0.33

Charpy Impact Testing

	Notch	Direction	Size	Temp °F	Position	Ft Lbf 1	Ft Lbf 2	Ft Lbf 3	Ft Lbf Avg	Shear 1	Shear 2	Shear 3	Shear Avg
Min Target	—	TRANS	FULL	32	—	15	15	15	20				
Actual	V	TRANS	FULL	32	TRAIL	320	325	322	322	100	100	100	100
Min Target	—	TRANS	FULL	-50	—	10	10	10	13				
Actual	V	TRANS	FULL	-50	TRAIL	241	266	253	253	100	100	100	100
Actual	V			-50	TRAIL	241	266	253	253	100	100	100	100

NDT Inspection: WMSEA

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star, without the expressed written consent of Vallourec Star negates the validity of the test report. Vallourec Star is not responsible for the inability of the material to meet specific applications.

Signed:

Name: PETER YANG
Date Approved: 11/28/2023

DRAFT

 Qualification Bend Report			
Customer:	DNOW	Report No.:	105093-QBR-1
Customer PO No.:	662682056	Code of Fabrication:	ASME B16.49-2023
TTB Job No.:	25H105093		

Mother Pipe Information

Pipe OD (in): 10.750 Pipe Grade: API-5L-X52 PSL 2 Pipe Heat No.: N90101

Bend Information

Radius (in): 50 Angle (degrees): 55 Bending Machine ID: M12
 Tangent 1 (in): 20 MIN Tangent 2 (in): 6 MIN Bend ID No.: 105093-QB-1

Essential Variables

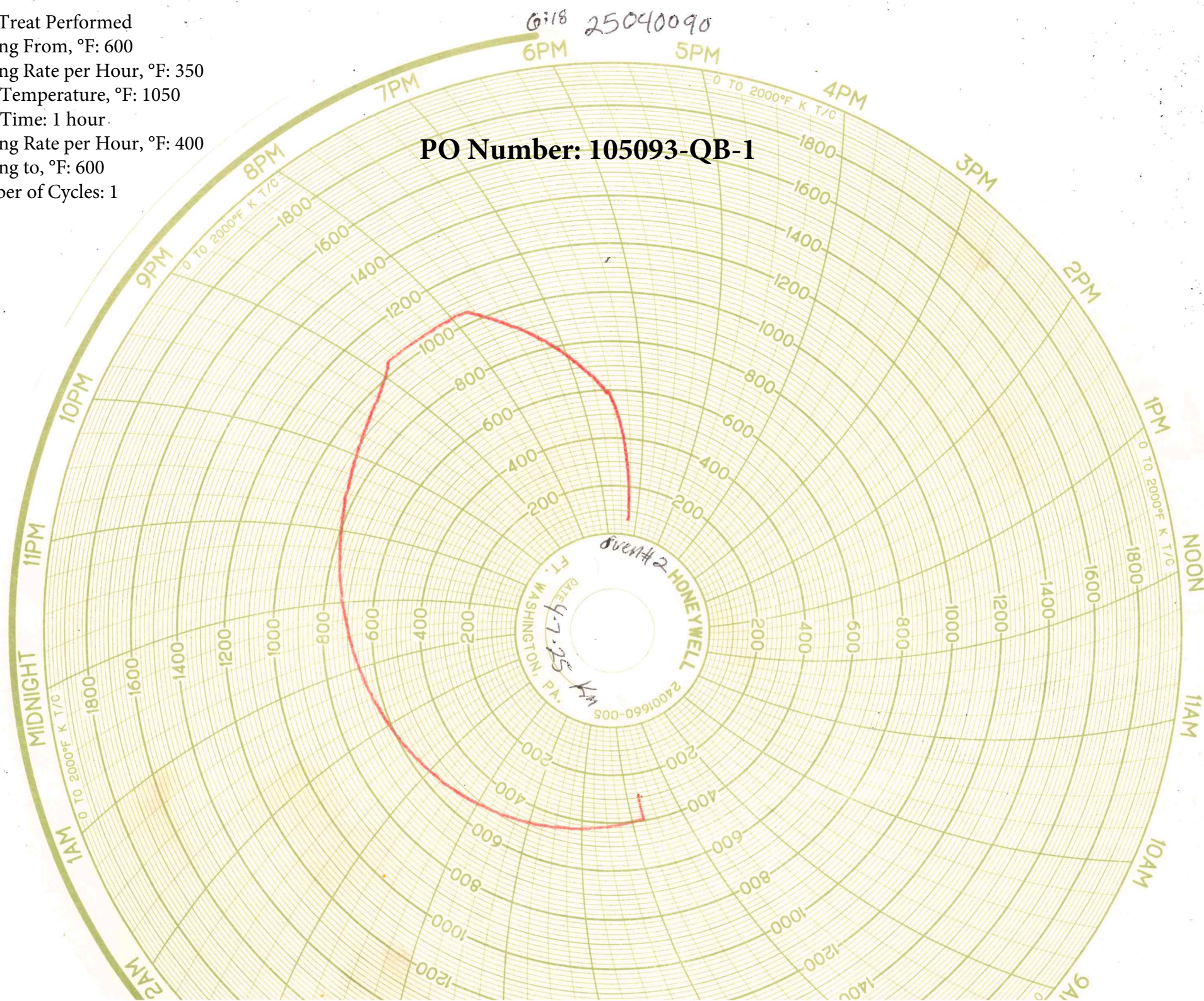
Essential Variable	Value	Limits of Variation
Carbon Equivalent	0.340	n/a
Pipe Diameter	10.750	None Allowed
Wall Thickness (inch):	0.500	+/- 0.12 inch
Radius-to-Diameter Ratio:	5.0	+1R - 0
Coil Design:	TTB High Spec	None Allowed
Coolant Type:	Water	None Allowed
Coolant Temperature (°F):	72.3	+/- 25°F
Coolant Flow Rate (GPM):	10.0	+/- 10%
Weld Seam Location:	n/a	15° from Neutral Axis
Induction Frequency (Hz):	871	+/- 20%
Heat Treatment Holding Temp (°F):	1050	+/- 25°F
Heat Treatment Soak Time (min):	60.0	+ 15 min / - 0 min

Note 1: Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

OPERATOR: MH **DATE:** 4/2/2025

Heat Treat Performed
Heating From, °F: 600
Heating Rate per Hour, °F: 350
Hold Temperature, °F: 1050
Hold Time: 1 hour
Cooling Rate per Hour, °F: 400
Cooling to, °F: 600
Number of Cycles: 1

PO Number: 105093-QB-1



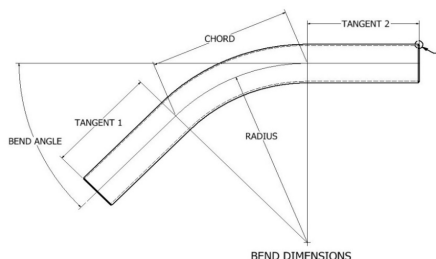


Dimensional-Visual Inspection Report

Customer:	DNOW	Pipe Size (in):	10.75 x 0.500	Report No.:	105093-QDV-1
Customer PO No.:	662682056	Grade:	API-5L-XS2 PSL 2	Bend ID No.:	105093-QB-1
TTB Job No.:	25H105093	Heat No.:	N90101	Code of Fabrication:	ASME B16.49-2023

Physical Dimensions

Dimension	Target	Tolerance	Actual
Chord (in)	46.2	n/a	46.2
Angle (deg)	55.0	+/- 0.5 °	55.0
Radius, R (in)	50	+/- 1%	50.0
Performed By:	MH	Date:	4/2/2025



Ovality $[(O.D. \text{ max} - O.D. \text{ min}) / O.D. \text{ nominal}] \times 100$

Quadrant	Limit	Min OD	Max OD	% Ovality
Start of Bend	3.0%	10.690	10.780	0.8%
1/4 of Bend Arc	3.0%	10.675	10.778	1.0%
1/2 of Bend Arc	3.0%	10.663	10.769	1.0%
3/4 of Bend Arc	3.0%	10.660	10.760	0.9%
Stop of Bend	3.0%	10.690	10.734	0.4%
Performed By:	MH	Date:	4/2/2025	

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.500	0.461	7.8%
1/4 of Bend Arc	10.0%	0.500	0.460	8.0%
1/2 of Bend Arc	10.0%	0.500	0.465	7.0%
3/4 of Bend Arc	10.0%	0.500	0.467	6.6%
Stop of Bend	10.0%	0.500	0.469	6.2%
Performed By:	MH	Date:	4/2/2025	

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	9.458 in
Gauge Length Required =	12" MIN		
Gauge passes through the entire bend without power equipment assistance			
Performed By:	MH	Date:	4/2/2025

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.			
Bend is stenciled with correct information.			
Performed By:	MH	Date:	4/2/2025

READINGS: ☒ MEET ACCEPTANCE CRITERIA
☐ DO NOT MEET ACCEPTANCE CRITERIA



Test Report

Attn: Firdosh Kavarana
Tulsa Tube Bending
4192 S Galveston Ave
Tulsa, OK 74107

Lab Number: 25040090
Date Received: 04/07/2025
Date Reported: 04/10/2025
PO Number: 105093-QB-1

Coupon Description: 10.75" OD x 0.500" Wall
Base Material 1: API-5L-X52 PSL 2, SMLS
Heat Number: N90101

Job Number: 105093

Heat Treat Performed

Heating From, °F: 600
Heating Rate per Hour, °F: 350
Hold Temperature, °F: 1050
Hold Time: 1 hour
Cooling Rate per Hour, °F: 400
Cooling to, °F: 600
Number of Cycles: 1

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -1C (Tangent)
Condition: After Heat Treat
Notch Type: V-Notch
Orientation: Transverse
Notch Location: Base Metal
Charpy Specimen Size: 10mm x 10mm
Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	240	73	80	80	78	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1, 2, and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.



Test Report

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -3C (STZ-Extrados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	184	180	211	80	82	89	84	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -4C (STZ-Intrados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	240	79	80	82	80	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1, 2, and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.



Test Report

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -5C (Bend Extradados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	240	77	73	76	75	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1, 2, and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -6C (Bend Intrados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	240	76	77	82	78	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1, 2, and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.



Test Report

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -8C (FTZ-Extrados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	232	77	80	72	76	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1 and 2 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.

Charpy V-Notch Impact Test per ASTM A370-24a / ASTM E23-23a

Specimen ID: -9C (FTZ-Intrados)

Condition: After Heat Treat

Notch Type: V-Notch

Orientation: Transverse

Notch Location: Base Metal

Charpy Specimen Size: 10mm x 10mm

Temperature, °F: +14

	Impact Value 1 (ft-lbf)			Lateral Expansion 1 (mils)				Shear 1 (%)			
	1	2	3	1	2	3	Avg	1	2	3	Avg
Results	240	240	240	81	82	78	80	100	100	100	100

Test Comments: Specified test requirements modified according to ASTM A370-24a, Table 10 when subsize specimens are permitted or necessary, or both.

Absorbed energy values above 80% of the 240 ft-lbf capacity of the impact tester are approximate. If the absorbed energy value(s) exceed(s) 192 ft-lbf, the average was not calculated per the requirements of ASTM E23-23a, Section 9.3.3.1.

Striker Radius: 8mm

Specimen numbers 1, 2, and 3 were unbroken as defined in ASTM A370-24a, Paragraph 26.4.3.7.



Test Report

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -1T (Tangent)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.481
Tensile Strength, psi	82200
Yield Strength at 0.5% EUL, psi	58900
Elongation After Fracture, %	33
YS/TS Ratio, %	0.72
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -3T (STZ-Extrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.504
Original Thickness, in	0.456
Tensile Strength, psi	83500
Yield Strength at 0.5% EUL, psi	59400
Elongation After Fracture, %	29
YS/TS Ratio, %	0.71
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"



Test Report

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -4T (STZ-Intrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.534
Tensile Strength, psi	81100
Yield Strength at 0.5% EUL, psi	56700
Elongation After Fracture, %	32
YS/TS Ratio, %	0.70
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -5T (Bend Extrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.462
Tensile Strength, psi	85600
Yield Strength at 0.5% EUL, psi	61000
Elongation After Fracture, %	26
YS/TS Ratio, %	0.71
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"



Test Report

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -6T (Bend Intrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.517
Tensile Strength, psi	84600
Yield Strength at 0.5% EUL, psi	58600
Elongation After Fracture, %	31
YS/TS Ratio, %	0.69
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -8T (FTZ-Extrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.464
Tensile Strength, psi	82700
Yield Strength at 0.5% EUL, psi	58500
Elongation After Fracture, %	28
YS/TS Ratio, %	0.71
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"



Test Report

Tensile Test per ASTM A370-24a / ASTM E8-22

Specimen ID: -9T (FTZ-Intrados)

Condition: After Heat Treat

Orientation: Transverse

	Results
Original Width, in	0.503
Original Thickness, in	0.517
Tensile Strength, psi	83700
Yield Strength at 0.5% EUL, psi	57000
Elongation After Fracture, %	32
YS/TS Ratio, %	0.68
Location of Fracture	Inside Middle Half of Gauge Length

Elongation: in 2"

Brinell Hardness Test, ASTM E10-23

Specimen ID: -1H (Tangent)

Condition: As Received

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	174

Brinell Hardness Test, ASTM E10-23

Specimen ID: -3H (STZ-Extrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	169



Test Report

Brinell Hardness Test, ASTM E10-23

Specimen ID: -4H (STZ-Intrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	174

Brinell Hardness Test, ASTM E10-23

Specimen ID: -5H (Bend Extrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	179

Brinell Hardness Test, ASTM E10-23

Specimen ID: -6H (Bend Intrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	179

Brinell Hardness Test, ASTM E10-23

Specimen ID: -8H (FTZ-Extrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	172



Test Report

Brinell Hardness Test, ASTM E10-23

Specimen ID: -9H (FTZ-Intrados)

Condition: After Heat Treat

Hardness Location: Surface

Hardness Test Method: ASTM E10-18 / ASTM A370-23

Hardness Scale / Test Load: HBW 10/3000 kgf

Measurement	Result
HBW	179

Approved By: Brent Fowler Title: Technical Director

Signature: BT Fh

*Test results relate only to the items tested. This document shall not be reproduced, except in full, without the written approval of American Piping Inspection, Inc. Metallurgical Laboratory.
Conformance statements are based on the Simple Acceptance Rule for measurement of uncertainty.